Verification supplement for the conditional Subset Sum refinement

Yanni Bills

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Abstract

We separate the obligations behind the published conditional exponent 0.4876723 and reconstruct their concrete mathematical interfaces. The supplement gives full written proofs of the filter-3 boundary transfer, analytic score bounds and integration error method, effective finite-circle approximation, and the isolation/preliminary-search case split. All 430 original continuous moment inequalities have now been validated and independently replayed using explicit inner/outer error bounds and ball arithmetic outside Lean. A separately reviewed Lean development establishes concrete boundary and regularity results, exact data links, real-logarithm and entropy bounds, all-row conditional transfers and unique region assignment. The remaining written links and end-to-end algorithmic obligations are distinguished from these results. The published release is preserved unchanged; no stronger unconditional running-time claim is made here.

1 Baseline, obligations and evidence levels

The baseline is public release `v1.0.0`, commit `5380ae2c6411f70769df6076ca82a54d43ff09c3` of `https://github.com/yannouli/subset-sum-conditional-refinement`. All additions live under `verification/`; the released manuscript, tuple, code, pins and historical evidence remain unchanged. The baseline assumes original algorithmic claims and 430 continuous moment inequalities. Publication, a Claimed listing, old Python checks and internal review are not substitutes for proofs of those assumptions.

The four groups are: (A) our changed boundary/transfer argument; (B) exact fixed data, real logarithm bounds and scalar allowances; (C) actual analytic scores, continuous moments and finite-circle limits; and (D) algorithmic correctness, precision and every-history word cost. The machine-readable obligation record is `theory/obligations.json`; intended statements are stated before implementation in `theory/intended-statements.md`. The accepted numerical evidence is described in Section 5. The separately accepted concrete Lean checkpoint is mapped precisely in Section 7; its limitations and the unfinished stronger replay are explicit.

Before and after the written reconstruction. The score bound and finite-circle approximation need not be retained as unexamined inherited assertions: Section 2 and Section 4 prove them for the concrete definitions. Section 8 similarly reconstructs the isolation and LD/preliminary-search implications. The moment inequalities are now supported by validated numerical evidence with explicit computational trust; the algorithmic dependencies in Section 9 remain separate. Validation of every continuous moment does not remove those remaining algorithmic obligations. No review badge is inferred.

2 The actual analytic functions

All integrals in this section use the probability average

$$\langle h\rangle = \frac{1}{2\pi}\int_0^{2\pi} h(t)\,dt.$$

In particular this is not an integral over a region selected by a pattern. For each of the 43 rational coefficient records of Kapila's Appendix A, put

$$r_v(x) = \sum_{j=1}^{8} r_{v,j}\cos(jx), \quad s_v(x) = \sum_{j=1}^{8} s_{v,j}\cos(jx), \quad F_v = \exp((\ln 2)r_v), \quad G_v = \exp((\ln 2)s_v).$$

Suppress v temporarily. The exponent $\kappa = \kappa_v$ is the exact source rational; $1 \le \kappa \le 2$. Define

$$\begin{aligned} Z_f &= \langle F\rangle, & Z_g &= \langle G\rangle, & Z_{f,\kappa} &= \langle F^\kappa\rangle, & Z_{g,\kappa} &= \langle G^\kappa\rangle,\\ f &= F/Z_f, & g &= G/Z_g, & f_* &= F^\kappa/Z_{f,\kappa}, & g_* &= G^\kappa/Z_{g,\kappa}. \end{aligned}$$

$$\begin{aligned} \mathcal{P}(x) &= \langle F(x+t)G(t)\rangle_t,\\ \mathcal{Y}(x) &= \langle F(x+t)G(t)(r(x+t)+s(t))\rangle_t,\\ Q(x) &= \mathcal{P}(x)/(Z_fZ_g),\\ K_x(t) &= f(x+t)g(t)/Q(x). \end{aligned}$$

The intended scores, following the definitions rather than their names, are

$$\begin{aligned} D(x) &= \langle K_x \log_2 K_x\rangle, & A_s(x) &= \langle K_x(t)\log_2 f_*(x+t)\rangle_t,\\ E_s(x) &= \langle K_x \log_2 g_*\rangle, & u(x) &= A_s(x)+E_s(x)-D(x), \quad w(x) = 2D(x)-A_s(x)-E_s(x). \end{aligned}$$

The subscripts distinguish scores A_s, E_s from a budget ratio or an exponent. All logarithms in scores are real base-two logarithms.

Lemma 1 (Concrete existence and score bound)**.** *For these definitions all denominators are strictly positive, the four density normalizations and K_x have average one, and all real scores are continuous and bounded. Every affine-certificate moment integrand is continuous and integrable. If each coefficient has absolute value at most two, then $|w(x)| \le 256$ for every x.*

Proof. The coefficient assumption gives $|r|, |s| \le 16$. Hence $2^{-16} \le F, G, Z_f, Z_g \le 2^{16}$ and $2^{-32} \le \mathcal{P} \le 2^{32}$. The tilted quantities satisfy the corresponding bounds with 16κ in place of 16. All assertions of positivity follow before taking any logarithm or quotient. Periodicity and substitution show that the averages of the normalized functions are one. The average of K_x is one by the definition of Q. Integration of continuous functions depending continuously on x on a compact rectangle preserves continuity; the strictly positive denominator bounds preserve it through division and logarithms. This proves integrability as well as continuity, and prevents a totalized logarithm or integral from masking a missing hypothesis.

Put

$$L = \log_2\mathcal{P} - \log_2(Z_fZ_g), \quad B = \mathcal{Y}/\mathcal{P} - \log_2(Z_fZ_g), \quad H = \log_2(Z_{f,\kappa}Z_{g,\kappa}) - \kappa\log_2(Z_fZ_g).$$

Here B is the K_x-average of $r(x+t)+s(t)-\log_2(Z_fZ_g)$. Substitution gives $D = B - L$ and $A_s + E_s = \kappa B - H$, so

$$u = L + (\kappa-1)B - H, \qquad w = (2-\kappa)B - 2L + H. \tag{1}$$

The quotient $\mathcal{Y}/\mathcal{P}$ is an average of $r+s$ with positive weights summing to one, hence has absolute value at most 32. Also $|\log_2(Z_f Z_g)| \le 32$ and $|\log_2 \mathcal{P}| \le 32$. Thus $|B|, |L| \le 64$. The tilted normalizer bounds give $|H| \le 64\kappa$. Since $0 \le 2-\kappa \le 1$,

$$|w| \le 64(2-\kappa) + 128 + 64\kappa = 256.$$

This derives the score bound through the actual definitions, rather than stopping at a bound on trigonometric coefficients. □

Lemma 2 (The required entropy order)**.** *The actual scores satisfy $D \ge 0$, $D - A_s \ge 0$, $D - E_s \ge 0$, and hence $w \ge 0$. The same is true for their exact finite-circle versions.*

Proof. If p, q are strictly positive normalized densities, the elementary inequality $\ln t \le t - 1$ gives

$$\langle p \ln(q/p) \rangle \le \langle q - p \rangle = 0.$$

Apply this with $p = K_x$ and $q = 1, f_*(x+\cdot), g_*$ respectively, and divide by $\ln 2 > 0$. Each q is normalized; for the shifted one this follows from periodic translation invariance. The same argument uses only normalized sums on the finite circle, where grid translations permute the summands. Finally $w = (D - A_s) + (D - E_s) \ge 0$. □

Independent normalization check. Eliminating all untilted normalizers from (1) gives the exact equivalent formulas

$$u = \log_2 \mathcal{P} + (\kappa - 1)\mathcal{Y}/\mathcal{P} - \log_2(Z_{f,\kappa} Z_{g,\kappa}), \tag{2}$$

$$w = (2-\kappa)\mathcal{Y}/\mathcal{P} - 2\log_2 \mathcal{P} + \log_2(Z_{f,\kappa} Z_{g,\kappa}). \tag{3}$$

The tilted normalizers do not cancel. These identities are useful for an independent implementation check, not alternative definitions that avoid the density correspondence.

3 A validated integration contract

This section proves a sufficient error bound. It does not assert that any numerical inequality has passed until the specified enclosures have actually been produced and checked. All source decimals are exact rationals, including coefficients and multipliers.

Lemma 3 (Periodic trapezoid bound)**.** *Suppose a 2π-periodic function h is holomorphic on a neighborhood of the closed strip $|\operatorname{Im} z| \le \sigma$, where $\sigma > 0$, and $|h| \le B_0$ there. For $N \ge 1$,*

$$\left| \frac{1}{N} \sum_{j=0}^{N-1} h(2\pi j/N) - \langle h \rangle \right| \le \mathcal{E}_N(B_0, \sigma) := \frac{2B_0}{e^{N\sigma} - 1}.$$

Proof. For a nonzero Fourier index k, shift the integral for its coefficient to $\operatorname{Im} z = -\sigma \operatorname{sgn} k$. Periodicity cancels the vertical contour sides and gives $|\widehat{h}(k)| \le B_0 e^{-|k|\sigma}$. Thus the Fourier series is absolutely convergent on the real axis. Averaging its terms at the N nodes kills every index not divisible by N. The error is at most $2B_0 \sum_{j \ge 1} e^{-jN\sigma}$, the claimed geometric sum. □

Inner integrals and arithmetic. Define $R(t) = \sum_j |r_j| \cosh(jt)$ and $S(t) = \sum_j |s_j| \cosh(jt)$. For an inner strip width $\sigma > 0$, $|r(z)| \le R(\sigma)$ and $|s(z)| \le S(\sigma)$. Lemma 3 therefore applies with the following strip bounds:

Inner quantity	Valid B_0
Z_f, Z_g	$2^{R(\sigma)},\ 2^{S(\sigma)}$
$Z_{f,\kappa}, Z_{g,\kappa}$	$2^{\kappa R(\sigma)},\ 2^{\kappa S(\sigma)}$
$\mathcal{P}(x_j)$	$2^{R(\sigma)+S(\sigma)}$
$\mathcal{Y}(x_j)$	$(R(\sigma)+S(\sigma))2^{R(\sigma)+S(\sigma)}$
$\langle 2^{2r_\beta} \rangle$	$2^{2\sum_j |r_j| \cosh(j\beta)\cosh(j\sigma)}$
$\langle 2^{2s_\beta} \rangle$	$2^{2\sum_j |s_j| \cosh(j\beta)\cosh(j\sigma)}$

Here $r_\beta(x) = \sum_j r_j \cosh(j\beta)\cos(jx)$; s_β is defined similarly. The node x_j is real, so the same convolution bounds hold uniformly for all outer nodes. Outward interval arithmetic must enclose the exact node sums, including the trigonometric arguments, exponentials, logarithms and all normalizers. Enlarge each inner interval by its own $\mathcal{E}_N$ before performing division or taking logarithms. Every denominator/logarithm interval used must have a verified strictly positive lower endpoint. A larger precision or agreement of two meshes is not a substitute for these error bounds.

The finite DFT acceleration is an exact sum identity. For node arrays $a_j = F(2\pi j/N)$ and $b_j = G(2\pi j/N)$ define $\widehat{a}_\ell = \sum_j a_j e^{-2\pi i \ell j/N}$ and use the inverse transform with prefactor $1/N$. Orthogonality of the finite geometric sum gives, with all indices read modulo N,

$$\mathrm{DFT}^{-1}(\widehat{a}_\ell \widehat{b}_{-\ell})_j = \sum_t a_{j+t} b_t.$$

Indeed, expansion of the left side retains exactly the terms with $-j_a + j_b + j = 0 \pmod N$. Thus the raw convolution node average is this inverse transform divided by N once more. For $\mathcal{Y}$, add the correlations of (Fr, G) and (F, Gs) and make the same division. The identity applies to exact node values. Rigorous ball DFT arithmetic must enclose that finite calculation; the analytic remainders above still have to be added afterward. An ordinary floating FFT without outward error control would not be a validated replacement.

Lemma 4 (A genuine zero-free strip)**.** *Let $\beta > 0$, and suppose rigorously computed bounds enclose*

$$A_f = \frac{\langle 2^{2r_\beta} \rangle}{Z_f^2}, \quad A_g = \frac{\langle 2^{2s_\beta} \rangle}{Z_g^2}, \quad q_0 = \sqrt{(A_f - 1)(A_g - 1)} < 1.$$

Then on $|\operatorname{Im} z| \le 2\beta$,

$$|Q(z) - 1| \le q_0, \qquad |L(z)| \le -\log_2(1 - q_0),$$

and Q has a holomorphic logarithm real on the real axis. Moreover

$$|B(z)| \le \frac{\sqrt{A_f A_g}(R(\beta) + S(\beta))}{1 - q_0} + |\log_2(Z_f Z_g)|. \tag{4}$$

Proof. The entire periodic functions f, g are real and even on the real circle because their defining trigonometric polynomials use only real cosines. They have shifted means one by contour shifting. For a function real on the real circle, its Fourier coefficients satisfy $\widehat{f}(-k) = \overline{\widehat{f}(k)}$. Parseval therefore writes the squared shifted L^2 norm as its constant term plus nonnegative multiples of $\cosh(2ky)$, so

it is nondecreasing in $|y|$. This also bounds both positive and negative imaginary shifts. At $y = \beta$ that squared norm is precisely A_f, since $|F(t + i\beta)|^2 = 2^{2r_\beta(t)}$. In particular $A_f, A_g \geq 1$.

For $z = x + iy$ with $|y| \leq 2\beta$, split the convolution contour:

$$Q(z) = \langle f(x + t + iy/2)g(t - iy/2)\rangle_t.$$

Subtract the product of the shifted means and apply Cauchy–Schwarz to $f - 1, g - 1$. Their squared norms are at most $A_f - 1, A_g - 1$; Here the centered identity is $\langle |f(\cdot + iy) - 1|^2\rangle = \langle |f(\cdot + iy)|^2\rangle - 2\,\mathrm{Re}\langle f(\cdot + iy)\rangle + 1 = \langle |f(\cdot + iy)|^2\rangle - 1$, and similarly for g. This proves the disk bound. The power series for $\mathrm{Log}(1 + h)$, with $h = Q - 1$, supplies the holomorphic branch and gives $|\log Q| \leq \sum_{j\geq 1} q_0^j/j = -\ln(1 - q_0)$. Divide by $\ln 2$ for the stated base-two bound. The strict inequality $q_0 < 1$ also supplies a neighborhood of the closed strip.

Apply the same split contour to $\mathcal{Y}/(Z_f Z_g)$. The absolute value of $r + s$ there is at most $R(\beta) + S(\beta)$; Cauchy–Schwarz bounds the integral of $|fg|$ by $\sqrt{A_f A_g}$. Since $|Q| \geq 1 - q_0$ and $B = \mathcal{Y}/(Z_f Z_g Q) - \log_2(Z_f Z_g)$, this gives (4). Real positivity alone would not justify the complex logarithm used in this proof. □

Proposition 5 (Per-certificate validated upper bound)**.** *For every filter occurring in a certificate or one of its comparison rows, suppose Lemma 4's bounds are enclosed independently. Let L_v^*, B_v^*, H_v^* bound $|L_v|, |B_v|, |H_v|$ on the strip, and put*

$$U_v^* = L_v^* + |\kappa_v - 1|B_v^* + H_v^*, \qquad W_v^* = |2 - \kappa_v|B_v^* + 2L_v^* + H_v^*.$$

For an old row vector $G_{v,i}$ form the following real-part upper bounds:

$$\begin{aligned}
\rho_0 :&\quad (\Delta_b + U_v^*,\ \Gamma_b + W_v^*,\ (U_v^* + U_j^*)_{j\in C}),\\
\rho_{1:8} :&\quad (A_b W_v^* + U_v^*,\ W_v^* - \Gamma_b L,\ \Gamma_b H + W_v^*,\ (U_v^* + U_j^*)_{j\in C}),\\
\rho_9 :&\quad (A_b W_v^* + U_v^*,\ W_v^* - 15\Gamma_b,\ (U_v^* + U_j^*)_{j\in C}).
\end{aligned}$$

Call the resulting vector $S_{v,i}$ and set

$$B_{v,i}^* = (2 + 2\cosh(2\beta))\, 2^{z_{v,i}\cdot S_{v,i}}.$$

If the interval $T_{v,i}$ encloses the average of the true outer node values after all inner/arithmetic errors, then

$$I_{v,i} \leq \sup T_{v,i} + \mathcal{E}_N(B_{v,i}^*, 2\beta).$$

Certification requires this upper bound to be strictly below an independently enclosed lower bound for $2^{\eta_b - m_i}$.

Proof. Each affine row's real part is bounded by the displayed entry; $A_b > 0$ and all multipliers are nonnegative. Hence the real part of the exponent is at most $z \cdot S$. The holomorphic extension uses $2^z = \exp((\ln 2)z)$ and the log branch already justified. Also $|2 + 2\cos z| \leq 2 + 2\cosh(2\beta)$. Lemma 3, followed by the enclosing node interval, gives the result. Negative entries such as $W_v^* - \Gamma_b L$ may be retained: they are valid upper bounds, not assumed nonnegative rows at complex points. □

Exact target and pilot. The targets needed by the released theorem are $m_0 = m_1 = .00025$, $m_2 = \cdots = m_8 = 4$, and $m_9 = 17$, with $\eta_b = .51613$. Kapila's common Proposition-B threshold alone is insufficient. The bounded pilot consists of $(3, 0), (3, 1), (4, 0)$ and at most one tail certificate, using the actual source comparisons for each. A separate record must classify every attempted pair as certified, unresolved or refuted. A failed sufficient bound is unresolved, not a refutation; a certified lower bound above threshold would be a refutation. If bounds are rigorous and the measured cost permits the finite full batch, the same contract can be applied to all 430 distinct pairs.

4 Finite-circle convergence and effective evaluation

Proposition 6 (All normalizers are part of the limit)**.** *Replace every inner average above by the normalized P-point average, including the tilted normalizers. At the grid points the resulting scores are precisely the finite-circle scores, and they converge uniformly to the continuous scores. The corresponding outer finite moments converge to their continuous moments. All convergence bounds can be made effective for this fixed finite coefficient family.*

Proof. For any continuously differentiable periodic real function h, comparison on each interval of length $2\pi/P$ with its left endpoint gives

$$|T_P h - \langle h \rangle| \le 2\pi \|h'\|_\infty / P.$$

The same estimate holds uniformly in an additional real parameter when its derivative bound is uniform. In this coefficient family, $|r'|, |s'| \le 2\sum_{j=1}^{8} j = 72$. Thus valid bounds on the integration-variable derivatives of FG and $FG(r+s)$ are respectively

$$144(\ln 2)2^{32}, \qquad 144(32\ln 2 + 1)2^{32}.$$

For F^κ, G^κ a bound is $144(\ln 2)2^{32}$. The raw normalizers and convolutions therefore differ from their continuous versions by explicitly bounded constants divided by P. Their positive lower bounds, for example $\mathcal{P}_P, \mathcal{P} \ge 2^{-32}$ and $Z_{f,P}, Z_f \ge 2^{-16}$, hold on either circle without a limiting argument. On these fixed positive compact intervals the quotient and logarithm operations are Lipschitz; for the latter use $|\log_2 x - \log_2 y| \le |x-y|/(\min(x,y)\ln 2)$. Substitution in (1) gives computable constants C_u, C_w such that the uniform score errors are at most $C_u/P, C_w/P$.

For a fixed certificate, the exponent error is at most a computable C_E/P. Its real exponents are uniformly bounded by another computable constant J. The mean-value bound for 2^x on this interval, and $0 \le 2+2\cos x \le 4$, bound the change in the node integrand by a constant divided by P. The continuous integrand is itself continuously differentiable with a computable derivative bound, so applying the Riemann estimate once more proves an explicit C_I/P moment error. All constants are finite expressions in the fixed rational coefficients, $\pi, \ln 2$ and elementary-function bounds, not input-dependent advice. Consequently a certified continuous upper bound separated from a desired finite threshold by $d > 0$ gives the effective choice $P > C_I/d$. There are finitely many modified certificates, so the maximum cutoff works for all of them, with any strictly smaller finite margin.

For a prescribed fixed accuracy, choose a rational mesh from these bounds and evaluate the continuous elementary functions by rational Taylor series with explicit remainders. Bounded positive logarithm arguments admit the convergent series for $\ln y = 2\sum_{j\ge 0}((y-1)/(y+1))^{2j+1}/(2j+1)$. Below the fixed cutoff compute finite averages directly; above it use the continuous approximation and the uniform error bound. Approximating the input residue angle uses polynomially many operations in $\log P$; all meshes and accuracy constants are fixed. This gives an effective uniform evaluation procedure, though not a claim that our Lean project has already formalized this analytic argument. □

These arguments reconstruct the analytic steps in the pinned OpenAI filtering/numerical sections and Kapila's finite-family adaptation [2, 1]. The proofs above supply their applicability to the actual 43-filter definitions; the numerical moment thresholds remain separate obligations until validated bounds are produced.

5 All 430 original moments: validated numerical results

The full batch now supplies a validated enclosure for every exact pair $(v, i) \in \{0, \dots, 42\} \times \{0, \dots, 9\}$, with no duplicate, missing or inferred pair. All 430 enclosures satisfy

$$I_{v,i} < 2^{.51613 - m_i}, \qquad m_0 = m_1 = .00025, \quad m_{2:8} = 4, \quad m_9 = 17. \tag{5}$$

The accepted artifact is `evidence/all-moments.json`, SHA256

`9646407eba82509f62c6464293d9fb472370d2e2c08a45ae07636c7bd0c84476`.

Its source hash is `b397531cc5997e88c439d5290774861b46a34aa32d9cd71fa7dcced130359298`, the unchanged pinned Kapila input. Each row preserves its node average, analytic outer error, enlarged integral enclosure, exact-threshold enclosure, strict gap and the needed inner/strip conditions.

The implementation uses Python 3.12.14 and `python-flint 0.8.0`, 160-bit ball arithmetic and one thread, with common half-width $\beta = .08$ and inner width $\sigma = .1$. The four-row pilot covered $(3, 0), (3, 1), (4, 0), (4, 9)$. The full run used at most 8192 outer nodes and completed in 36.81 seconds, within its 600-second and 131072-node ceilings. These are validation-batch times, not subset-sum solver times. The numerical code was frozen in implementation commits `398e7aa`/`f36b062`; the latter also preserves direct-sum checks of the DFT convention.

For scale, the following deliberately outward rounded bounds summarize the pilot pairs as present in the full artifact. They are not a substitute for its other 426 separate certificates.

Pair (v, i)	Nodes	Integral upper bound	Threshold lower bound
$(3, 0)$	4096	1.385921	1.429866
$(3, 1)$	512	1.426917	1.429866
$(4, 0)$	1024	1.425608	1.429866
$(4, 9)$	2048	$3 \cdot 10^{-37}$	$1.09109 \cdot 10^{-5}$

Independent replay and trust boundary. Procopius independently replayed every pair in a separate Python environment in 36.14 seconds. Every numerical enclosure, threshold, gap and node count matched; timing metadata naturally differed. The review also checked exact Cartesian pair coverage, coefficient and row correspondence, nonnegative multipliers, both tilted normalizers, positive inner denominators, all comparison filters on the common zero-free strip, and the explicit inner/outer remainders. Direct finite-sum tests checked the library DFT sign and normalization; separate missing, duplicate and altered source-record copies were rejected. The numerical review is recorded in review commit `7f1e398`, receipt `47a33c19`.

These are rigorous numerical results conditional on the correctness of the source parser, Python bindings, FLINT/Arb ball elementary functions and DFT implementation, together with the written analytic error theorem proved here. They are not proofs checked inside Lean. The enclosures include discretization and rounding errors; neither mesh agreement nor a heuristic integration error estimate is their justification. Internal agent review is not independent human peer review or a site-awarded status.

The portable commands and exact environment are in `moments/README.md`. From the repository root, the documented pinned fetch places the input under `verification/evidence/`; `verification/moments/validate.py` with `-all` regenerates all certificates. Use a new output filename when replaying so the accepted artifact remains unchanged. Certificate evaluation is independent of parameter discovery; no optimization is rerun.

Assumption change. Before this validation, all 430 inequalities in (5) were separate inherited assumptions. They may now be replaced by the accepted validated evidence under the computational trust just stated. The precise thresholds needed by the released theorem are covered, including the stronger middle and tail margins. No alias-coverage, checksum-estimator or complete implementation theorem follows merely from those integral bounds.

6 Reconstructing the changed certificate

This proof starts from the two exact source records, not from the released note's conclusion. Use zero-based filter indices and the actual functions of Section 2. Kapila's Appendix C gives

$$\begin{aligned}
E^b_{3,0}(x) &= 17.211(\Delta_b - u_3(x)) + 27.1977(\Gamma_b - w_3(x)) \\
&\quad + 8.3302(u_3(x) - u_5(x)) + 3.0053(u_3(x) - u_{16}(x)), \\
E^b_{3,1}(x) &= 7.7547(A_b w_3(x) - u_3(x)) + 0(w_3(x) - \Gamma_b) + 0(1.25\Gamma_b - w_3(x)) \\
&\quad + 4.2699(u_3(x) - u_5(x)) + 2.2635(u_3(x) - u_8(x)) \\
&\quad + .461(u_3(x) - u_{16}(x)) + 3.4032(u_3(x) - u_{35}(x)).
\end{aligned}$$

Here $\Delta_b = .02844$, $\Gamma_b = .45925$, $A_b = \Delta_b/\Gamma_b$. In particular both zero entries are exact rationals; they are not small floating-point estimates. The complete finite-data authentication must identify exactly the expected pairs $(v, i) \in \{0, \dots, 42\} \times \{0, \dots, 9\}$ and these comparison orders, not only a row count.

For the fixed published tuple put

$$\Delta = \eta - \tau, \quad \Gamma = 2\tau - \eta, \quad A = \Delta/\Gamma, \quad q = \Gamma/\Gamma_b, \quad K = \Delta/q - \Delta_b.$$

The numerical obligations include $\Delta > 0$, $0 < \Gamma < \Gamma_b$, $K > 0$, so $A - A_b = K/\Gamma_b > 0$. For pattern values U_j, W_j select filter 3 only when $U_3 \geq U_j$ for every j. Set $U = U_3, W = W_3$ and $\alpha = 1$ for $W \leq \Gamma$, $\alpha = \Gamma/W$ otherwise. Only $\alpha U < \Delta$ is the sampled case. Define $E'_{3,0}$ by replacing the first two constants in $E^b_{3,0}$ by Δ, Γ. Define $E'_{3,1}$ by replacing A_b by A and replacing the zero upper-row multiplier by $7.7547(A - A_b)$; the lower-row multiplier remains exactly zero.

Theorem 7 (Concrete domination and integral identities)**.** *On sampled selected-filter-3 patterns, use $E'_{3,0}$ for $W \leq \Gamma$ and $E'_{3,1}$ for $\Gamma < W \leq 1.25\Gamma_b$. Then, with exponents evaluated at the pattern values,*

$$-\alpha U \leq -\Delta + {E'_{3,i}}^{\pi}.$$

For the actual analytic functions and the full normalized circle,

$$\begin{aligned}
\langle (2 + 2\cos x) 2^{E'_{3,0}(x)} \rangle &= 2^{\Lambda_0} \langle (2 + 2\cos x) 2^{E^b_{3,0}(x)} \rangle, \\
\langle (2 + 2\cos x) 2^{E'_{3,1}(x)} \rangle &= 2^{\Lambda_1} \langle (2 + 2\cos x) 2^{E^b_{3,1}(x)} \rangle,
\end{aligned}$$

where

$$\Lambda_0 = 17.211(\Delta - \Delta_b) + 27.1977(\Gamma - \Gamma_b), \qquad \Lambda_1 = 7.7547(1.25)K.$$

Proof. In the low case $\alpha = 1$, $\Delta - U > 0$, $\Gamma - W \geq 0$, and all listed comparisons are nonnegative. Since $17.211 \geq 1$, ${E'_{3,0}}^{\pi} \geq \Delta - U$. For the extended first band, $W > \Gamma > 0$ implies $0 < \alpha < 1$, $\alpha AW = \Delta$, and $AW - U > 0$. The new upper multiplier is nonnegative and its row $1.25\Gamma_b - W$ is nonnegative. The comparison rows are nonnegative. Therefore

$${E'_{3,1}}^{\pi} \geq 7.7547(AW - U) \geq \alpha(AW - U) = \Delta - \alpha U.$$

The lower row $W - \Gamma_b$ may be negative. Its contribution is zero because its multiplier is exactly zero; no lower-boundary assumption was silently added to this proof.

For every real x, direct subtraction gives $E'_{3,0} - E^b_{3,0} = \Lambda_0$. In the first band the changed terms are

$$7.7547(A - A_b)w_3(x) + 7.7547(A - A_b)(1.25\Gamma_b - w_3(x)) = 7.7547(A - A_b)1.25\Gamma_b = \Lambda_1.$$

These are identities at every point of the circle, including those unrelated to any selected pattern. Lemma 1 supplies actual continuity/integrability. The identity $2^{a+b} = 2^a 2^b$ and linearity of the normalized integral permit pulling the constant factor out. No integration domain was shortened or selected by a pattern condition. □

Lemma 8 (Ordinary transfer and complete coverage)**.** *For every other low record use $\Delta/q - u_v$ in place of $\Delta_b - u_v$. For every bounded old band $\Gamma_b L < W \le \Gamma_b H$, replace A_b by A and add $z_0(A - A_b)$ to its upper-row multiplier. For the tail make only the change $A_b \mapsto A$. These give sampled-pattern domination, with pointwise shifts $z_0 K$, $z_0 KH$, and $z_0(A - A_b)w_v(x)$ respectively. For filter 3 use Theorem 7 in its first two regions. Together the regions are disjoint and exhaustive, with every finite upper endpoint assigned to the preceding region.*

Proof. For an ordinary low pattern $W \le \Gamma_b$, $q \le \alpha \le 1$. If $U \ge 0$, sampling and $\alpha \ge q$ imply $\alpha U \le U$ and $U < \Delta/q$; more directly in either sign case $\alpha U < \Delta \le \alpha\Delta/q$, so $\Delta/q - U > 0$. Also $\alpha\Delta/q \ge \Delta$. Since $z_0 \ge 1 \ge \alpha$, $z_0(\Delta/q - U) \ge \Delta - \alpha U$. The remaining rows are nonnegative with nonnegative weights. The exponent difference is $z_0 K$.

In a bounded old band, $W > \Gamma_b L \ge \Gamma_b > \Gamma$ and $\alpha < 1/L \le z_0$. Again $\alpha AW = \Delta$ and $AW - U > 0$. The old lower and upper rows are nonnegative; the upper multiplier only increases. Domination follows. The $w_v(x)$ terms cancel in the exponent difference, leaving $z_0(A - A_b)\Gamma_b H = z_0 KH$. The tail proof is identical except for the missing upper row. Its shift is bounded above by $256 z_0(A - A_b)$ using Lemma 1.

The ordered old endpoints are $\Gamma_b, 1.25\Gamma_b, 1.6\Gamma_b, 2\Gamma_b, 2.6\Gamma_b, 3.5\Gamma_b, 5\Gamma_b, 8\Gamma_b, 15\Gamma_b$. Replacing only filter 3's first endpoint by Γ preserves order because $0 < \Gamma < \Gamma_b < 1.25\Gamma_b$. An ordered list partitions the real line into the initial closed ray, successive intervals open on the left and closed on the right, and a final open ray. Thus $W = \Gamma$ belongs to the low case, while $W = 1.25\Gamma_b$ belongs to the extended first band; no gap or overlap occurs. All other endpoints and record assignments are unchanged. □

Connection to the fixed certificate. For a nonnegative old weight vector of sum at most $Z = 418.36$, the bounded-band update has sum at most $Z(1 + A - A_b) < 419$ once the fixed scalar inequality has been proved. Other updates do not increase the sum. The source maxima required for the five moment families are 10.6428 for the 42 ordinary low rows and 8.4537 for the first bands; the two special filter-3 rows have the shifts just proved. Every row's pattern-to-coordinate-average error is at most 6ϵ when $A < 1/4$: a first band row has error at most $(4A + 3)\epsilon < 4\epsilon$, a comparison has at most 6ϵ, and low/boundary rows have at most 4ϵ. Constants contribute no error and the zero row contributes zero. This yields the released allowance $6Z'\epsilon$. The 18ϵ estimator allowance remains an algorithmic dependency, not something implied by this row-error calculation.

The identities above transfer each old full-circle moment with its own record. The fixed-data margin theorem must connect every one of those records to the corresponding proved transfer; a generic lemma and an unconnected table check would not establish that conjunction.

7 Accepted concrete Lean checkpoints

The completed checkpoint is implementation commit `7f9548e`, with target `Verification.DataBridge` importing `Concrete`, `Boundary` and `Regularity`. It uses Lean 4.24.0 with the following dependency pins:

> Wharton: `bed8e18f96db5955c1424abf3005b4fed4bd8d77`
> mathlib: `f897ebcf72cd16f89ab4577d0c826cd14afaafc7`.

The full lockfile retains the other dependency pins. The accepted theorem-to-manuscript mapping is deliberately scoped:

1. **Pattern domination.** `Verification.Published.low_pattern_domination` and `Verification.Published.extended_pattern_domination` establish Theorem 7's two domination inequalities for the actual filter-3 records and published budgets. Their explicit hypotheses are the relevant interval bounds, the listed nonnegative comparisons and the sampled-pattern inequality; the desired domination is not assumed. The exact zero lower multiplier is proved and used.

2. **Actual normalized-circle moments.** `Verification.Published.concrete_low_moment_identity` and `Verification.Published.concrete_band_moment_identity` prove the full-circle shift conclusions of Theorem 7, together with integrability of both actual integrands and strict positivity of the old moment. The definitions are the original entropy-score integrals, not arbitrary functions called moments. The measure is Lebesgue measure restricted to $[0, 2\pi]$, with the prefactor $(2\pi)^{-1}$ in the average. The positive-support proof is inside that full interval. Continuity and positive denominators are proved in `Regularity`, rather than assumed to avoid a totalized-integral or totalized-logarithm artifact.

3. **Data bridges.** `filter_transcription`, `certificate_transcription`, `exact_pair_coverage`, `source_filter_dimensions`, `rawFilter_index_in_bounds` and `comparison_zip_no_truncation` in the same namespace check the exact ordered 43-filter/430-record data and their dimensions. The other bridge theorems give exact list lengths, comparison uniqueness and the ordinary-low cap excluding filter 3. The literal data equalities are kernel checked; correspondence of those literals to the authenticated external TeX tables was also independently parsed and compared.

4. **Limited interval result.** `Verification.Published.complete_region_coverage` proves the disjunction of the ten modified filter-3 intervals, with the indicated open/closed endpoints. At this checkpoint it does *not* prove uniqueness, disjointness, or a region assignment for every ordinary filter. Those broader claims have the written proof in Lemma 8, but their subsequent formal certificate links are not credited to this checkpoint.

Build, axioms and separate checks. The target and audit built from source, and Procopius independently replayed all four completed modules using the pinned built imports. The separately rerun transitive axiom report contains only `propext`, `Classical.choice` and `Quot.sound`. No accepted proof uses `sorry`, native-evaluation trust shortcuts or a custom mathematical axiom supplying missing conclusions. Explicit mathematical hypotheses were inspected separately from the axiom report. The source/statement/axiom receipt is under `evidence/boundary-checkpoint/`; independent acceptance is review commit `097e859`, receipt `30e28528`.

The stronger all-import `lean4checker -fresh` attempt exceeded its five-minute cap and is *unfinished*, not passed. Successful module source replay using built dependencies does not replace

that stronger imported-environment recheck. A separate automated Comparator statement check was not run; the pinned Lake provides no such command. Actual definitions and statements received independent meaning review. No official challenge statement, website check or status upgrade is claimed. The exact replay commands, elapsed times, hashes and timeout are preserved in the checkpoint receipt.

Accepted fixed-certificate extension. Implementation commit `605cf983` adds `LogScalar`, `Scalar`, `RowMargins`, `Regions` and `AllRows` to target `Verification`. The combined source build and `AuditScalar.lean` pass. Procopius independently replayed all five added modules and checked the statements and permitted axioms: review commit `bd3cf784`, receipt `e3f55627`. Its accepted correspondence is as follows. The three `entropy_a`, `entropy_twoa` and `entropy_hma` statements concern actual real logarithms and binary entropy, not merely inequalities between supplied interval endpoints. `all_alias_gaps` checks the twenty fixed alias and auxiliary gaps. `all_430_transferred_margins` and `all_430_weight_sums` link the actual records to the transfer allowances and rounding budget. `full_region_partition` states existence and uniqueness of region assignment for every filter, including the boundary conventions. These broader results are distinct from the earlier filter-3 interval disjunction.

The analytic link is `all_430_conditional_moment_bounds`: for each imported record, it transfers the actual normalized-circle old moment to the new threshold. Its explicit premises include the old moment inequality and $w_v(x) \le 256$ for every real x. We discharge these respectively with the accepted outside-Lean certificates in Section 5 and the written score proof in Section 2; neither is a Lean conclusion of this theorem. The reduced-score identities used by the numerical validation also retain their written proofs. Moreover, per-record analytic exponent/moment transfer is distinct from assembling sampled-pattern domination in every ordinary region: the latter remains written, while the new filter-3 pattern inequalities are already formalized.

Finally, `exact_error_and_display_gap` and `fixed_tuple_ordinary_O` concern the fixed work exponent 0.48767228002, its strict allowance below 0.4876723, and absorption of the stated sublinear and logarithmic overhead. They do not prove that the full algorithm satisfies that resource premise. The 430 moment thresholds remain outside-Lean validated results, and the full algorithmic obligations in Section 9 remain open. This extension retains the unfinished fresh-replay and separate statement-check limitations described above.

8 Two inherited algorithmic obligations reconstructed

The following local results have complete written proofs. They do not settle the separate alias-estimator or finite-precision arguments.

Theorem 9 (Transformation and isolated bin)**.** *Let a_i be positive integers and $t \ge 0$. Choose a power of two $N_0 \in [12n, 24n)$, independent uniform $u_i \in \{1, \dots, N_0\}$, and put $C = 1 + nN_0$, $c_i = Ca_i + u_i$, $t_s = Ct + s$ for $0 \le s \le nN_0$. For every choice of the tie weights,*

$$\exists I \subseteq [n] : a(I) = t \quad \Longleftrightarrow \quad \exists s, I : c(I) = t_s.$$

On a NO input every transformed bin is empty. On a YES input the probability that at least one transformed bin has exactly one index-set representation is at least $11/12$.

Proof. If $a(I) = t$, take $s = u(I) \in [0, nN_0]$. Conversely, $c(I) = Ct + s$ implies $C(a(I) - t) = s - u(I)$, whose absolute value is at most $nN_0 < C$. Since $a(I) - t$ is an integer, it is zero.

For completeness, let $\mathcal{F}$ be any nonempty family of subsets of $[n]$, here the original solution family. For each coordinate i, fix all u_j, $j \neq i$. If both classes exist, let m_0 be the least tie sum among members not containing i and m_1 the least tie sum excluding u_i among members containing i. Equality of the two class minima requires $u_i = m_0 - m_1$, at most one of its N_0 possible values. If either class is empty there is no such event. If the overall minimum is attained by two different subsets, some coordinate distinguishes them, so that coordinate's class minima are equal. A union bound, conditioning on the other weights for each coordinate, gives probability at most $n/N_0 \leq 1/12$ of a nonunique minimum. A unique minimum solution has a unique tie sum within the solution family, and hence a unique transformed bin by the equivalence already proved. Nothing excludes repeated a_i or the empty subset. □

Lemma 10 (Deficiency on uniform half-blocks)**.** *For fixed transformed weights define*

$$\Sigma_c(V) = \{c(I) : I \subseteq V\}, \qquad d_c(V) = |V| - \log_2 |\Sigma_c(V)|.$$

Inserting one index changes deficiency by a value in $[0, 1]$*. Replacing one index by another changes it by at most one in absolute value. For a uniformly sampled* k*-subset* V_k*,*

$$\mathrm{Var}(d_c(V_k)) \leq k/4 \leq n/4.$$

Also $f(k) = \mathbb{E}d_c(V_k)$ *is nondecreasing in* k*.*

Proof. For $i \notin V$, $\Sigma_c(V \cup \{i\}) = \Sigma_c(V) \cup (c_i + \Sigma_c(V))$. Its cardinality lies between $|\Sigma_c(V)|$ and $2|\Sigma_c(V)|$, proving the insertion claim. Two sets differing by an exchange both arise by inserting an index into their common part, so their deficiencies lie in an interval of length one. Sampling a uniform $(k+1)$-subset and deleting a uniform member couples the uniform sizes $k+1, k$; monotonicity follows.

Expose the first k members $X_1, \ldots, X_k$ of a uniformly random ordered sample without replacement, and let M_j be the conditional expectation of the final deficiency after j exposures. At a fixed prefix, compare next choices a, b. Transposing a, b gives a bijection between the possible remaining completions. The corresponding final sets are either identical (both indices eventually occur) or differ by one exchange. Hence the two conditional expectations differ by at most one. Conditional on the prefix, M_j therefore lies in an interval of length at most one. A real random variable in an interval of length one has variance at most $1/4$: expand $(X - \min X)(\max X - X) \geq 0$ and complete the square in its mean. Thus each martingale difference has conditional second moment at most $1/4$. Distinct differences are orthogonal by iterated conditional expectation. Summing their variances gives the bound $k/4$. □

Theorem 11 (A capped preliminary search and the LD split)**.** *Let* $n \geq 2$*,* $b \geq 1$*, and use the fixed published* $0 < \tau < 1/2$*. Fix the tie weights and* $\zeta > 0$*. The condition*

$$\mathbb{E}d_c(V_{\lfloor n/2 \rfloor}) \leq (\tfrac{1}{2} - \tau + \zeta)n \tag{LD}$$

is exactly the low-deficiency condition used here. There is a preliminary procedure returning an exact decision or CONTINUE, using $O(2^{\tau n}\,\mathrm{poly}(n + b + 2))$ *word operations on every history. If LD fails, for all sufficiently large* n *it returns CONTINUE with probability at most*

$$\frac{2}{\zeta^2 n} + n2^{-n^2}.$$

This conclusion concerns a fixed-input probability; an observed CONTINUE is not a certificate that LD holds.

Proof. Sample a uniform half-block V and its complement, of sizes $\lfloor n/2 \rfloor$ and $\lceil n/2 \rceil$. If LD fails, either block U has expected deficiency greater than $(\frac{1}{2} - \tau + \zeta)n$, by Lemma 10. Chebyshev's inequality with variance at most $n/4$ gives

$$\Pr\{d_c(U) < (\tfrac{1}{2} - \tau + \zeta/2)n\} \leq \frac{1}{\zeta^2 n}.$$

Outside these two bad events,

$$\log_2 |\Sigma_c(U)| = |U| - d_c(U) \leq \tau n - \zeta n/2 + 1/2 < \tau n$$

when $n > 1/\zeta$. Both lists of distinct half sums fit within $B_n = 2^{\lceil \tau n \rceil}$.

Construct such a sorted list by starting with $\{0\}$ and, for each weight, merging the current sorted list with its translate, deleting duplicate values. If the next distinct output would exceed B_n, return CONTINUE before writing it. The number of distinct sums is nondecreasing under insertions, so if the final list fits, every prefix fits. Each merge costs $O(B_n)$ operations. Once both full lists fit, the sorted two-pointer search answers exactly whether some left and right sums total any required t_s. There are $O(n^2)$ targets, so even checking them separately costs only a polynomial times B_n. If either list hits its cap, no inexact decision is emitted.

For an explicit bounded sampler, expose a uniform ordered sample without replacement using partial Fisher–Yates swaps. A required integer in $\{0, \ldots, m-1\}$ is obtained by masking a fresh uniform word to $\lceil \log_2 m \rceil$ bits and rejecting values at least m. Acceptance is at least $1/2$. Cap each draw at n^2 proposals and return CONTINUE on exhaustion. Couple this with the ideal unbounded sampler: at most n draws are made, so discrepancy probability is at most $n2^{-n^2}$. The ideal subset is exactly uniform. The bounded sampler has polynomial cost on every history.

All sums have $O(b+\log n)$ bits: $N_0 < 24n$ and $C < 24n^2+1$ give $\sum_i c_i < 31n^3 2^b$ for $n \geq 2$, $b \geq 1$. The bit bound $b+3\log_2 n+5$ is below the prescribed $\lceil 4(n+b+\log_2(n+2)) \rceil$ word size on this domain, so sums and keys occupy a fixed number of words. Two capped final lists plus the alternating merge workspace can be kept in at most $4(B_n+1)$ words: the translated list is read implicitly, not copied. The partial Fisher–Yates array, input and counters require at most Kn additional words for a fixed implementation constant K. Thus simultaneous storage is at most $4(B_n+1)+Kn$, whose address length is $\tau n + O(\log n)$. Since $\tau < 1/2$, this is below the prescribed word size for all n above a fixed implementation cutoff. For the finitely many smaller $n \geq 2$, enumerate subsets with a running sum in $O(n)$ storage and answer exactly; its 2^n time contributes only a fixed factor to the stated $O(2^{\tau n} \operatorname{poly}(n+b+2))$ bound. This cutoff is independent of the input bit length and of c. Fixed rational τ permits exact calculation of its ceiling. Every allocation and loop has an explicit cap, including the failure histories, proving the stated all-history cost. □

These proofs reconstruct the transformation and preliminary-search results of the pinned OpenAI setup and Kapila Sections 2.1/4 [2, 1]. They establish the local statements under their displayed hypotheses without assuming an exact subset-sum solver or an LD oracle. In the composed analysis one fixes the tie weights first, splits according to LD, and draws the main procedure's independent randomness whether or not an earlier exact answer makes it unused. This avoids conditioning the main proof on the observed continuation.

9 The consequential algorithmic dependencies that remain

Proposition 12 (Exact alias classes and sound insertion)**.** *Use the actual integer key $(c(A), b(A))$ for each half-block subset, where $b_i = c_i \bmod P$ is the least nonnegative residue. Keep a dictionary*

key (κ_L, κ_R, Z) only once, and store the value-map phase sums Φ_L, Φ_R over all subsets with the respective key. Every insertion made by Kapila's Section 5.2 matching/expansion rule is an alias class. If all alias classes are reached, the exact output equals the full alias phase sum, with no multiplicity error from repeated discoveries or repeated original weights.

Proof. For precision, $T_{s,q} = (t_s \bmod P) + qP$, $0 \le q < n$. Every modular representation has exactly one such carry: its integer sum of reduced weights is in $[0, n(P-1)] \subset [0, nP)$ and is congruent to the least residue of t_s. Subtracting that residue and dividing by P gives an integer in $\{0, \dots, n-1\}$. Write $c_i^F = (-1)^{[i \in F]} c_i$ on the large block, and similarly for b^F. Directly partitioning Z into $Z \setminus F$ and $Z \cap F$ proves $c^F(Z) = c(Z \triangle F) - c(F)$, and the same identity for b. In an insertion the masks X, Y are disjoint; put $Z = (X \cup Y) \triangle F$. The integer matching condition is

$$b(\kappa_L) + b^F(X) + b(\kappa_R) + b^F(Y) = T_{s,q} - b(F),$$

where $T_{s,q} \equiv t_s \pmod P$. The sign identity gives $b(A \cup Z \cup B) = T_{s,q}$ for every half-block pair having those keys, and thus its unreduced sum is congruent to t_s modulo P. The skipped exact interval in the matching rule excludes precisely

$$c(\kappa_L) + c^F(X) + c(\kappa_R) + c^F(Y) = t_s - c(F).$$

The same identity therefore shows that each inserted member has unreduced sum different from t_s. It is an alias, not an exact representation.

Every index set I belongs to exactly one class: $(\kappa(I \cap L), \kappa(I \cap R), I \cap M)$, for the fixed disjoint partition L, M, R. Every member of a class has the same unreduced and reduced sums. Additivity of the retained phases gives

$$\sum_{I \text{ in class}} \chi(I) = \left(\sum_{A : \kappa(A) = \kappa_L} \chi(A) \right) \left(\sum_{B : \kappa(B) = \kappa_R} \chi(B) \right) \chi(Z) = \Phi_L(\kappa_L) \Phi_R(\kappa_R) \chi(Z).$$

Distinct class keys partition the relevant index sets, and the dictionary stores each key at most once. Consequently reaching every alias class suffices for the claimed exact sum. This does not prove that the randomized procedure reaches every class before its cap; that coverage/cost/probability obligation remains below. □

The following are exact obligations, not conclusions of the local proofs or of scalar certificate checking.

1. **Alias coverage and cost.** Under LD, with the published block sizes and slack, Kapila's Section 5 procedure must return every modular alias contribution in its deduplicated value classes with the retained signing, except with a uniformly small failure probability. Its expected-size estimates must imply the stated deterministic caps through the actual abort policy, without losing the per-input success bound. The half-block bad-prime bound uses $\delta = (\eta - (\mu/2)\log_2 3)/2$; the actual requirements include $\delta > 1/2 - \tau + \zeta$ and the bit-length guard slack $\theta < \delta$.

2. **Same-phase cancellation.** For each target and modular prime, the alias sum and the checksum must use identical stored phases. Every modular representation with a different integer sum must occur exactly once in the deduplicated class sum. This gives exact residual zero on NO inputs and a modulus-one residual at an isolated bin; it does not by itself bound either numerical estimator's error.

3. **Block-table estimator.** For the actual finite-circle density functions, every selected sampled pattern must satisfy the table-coverage/second-moment hypotheses in OpenAI Section 4 as adapted by Kapila Section 6. The released 18ϵ allowance belongs here. The distribution used to factor coordinate moments must be the original independent phase law, not a law conditioned on a favorable pattern, table event, alias-success event or CONTINUE.

4. **Finite precision and uniform constants.** Scores and elementary functions must be evaluated with enough accuracy for the actual block thresholds. Stored phases must remain consistent across both subroutines; all accumulated errors must stay below the decision threshold gap. Section 4 supplies effective fixed-score evaluation, but not the full propagated complex-arithmetic error bound. The program must generate required constants/tables, not receive them as advice.

5. **Every-history resources.** Prime search, table construction, trial lists and visits, rational arithmetic, random draws, memory and indirect addresses all require deterministic caps with charged word cost. A finite expected work estimate alone is insufficient. The published scalar gaps verify proposed exponents, not that a described procedure actually obeys each one. The guarded program and exact fallback must be a single program independent of the polynomial-bit parameter c.

Precise cancellation fact. If $\chi(I)$ is the same retained unit-modulus phase in both sums, then, identically,

$$\sum_{c(I)\equiv t_s \pmod P} \chi(I) - \sum_{\substack{c(I)\equiv t_s \pmod P \\ c(I)\neq t_s}} \chi(I) = \sum_{c(I)=t_s} \chi(I).$$

This is a finite partition identity, valid with repeated original weights because subsets are index sets. It proves the cancellation *if* the two enumerated/sampled quantities correspond to those exact sums with common phases. Completeness of the class algorithm and accuracy of the estimates remain separate obligations; renaming their outputs by these sums would assume the desired conclusion.

The current supplement therefore does not claim that the complete word-RAM algorithm has been independently verified. Any unresolved point above remains an explicit inherited dependency even after the boundary, scalar and continuous-moment obligations are settled.

9.1 A bounded precision obligation that can be discharged locally

Proposition 13 (Alias-sum arithmetic, given the correct dictionary)**.** *Suppose the phase maps and dictionary are those of Proposition 12, with coverage already supplied. Use the same exact retained phase vector modulo P throughout. Compute each unit phase to absolute complex error at most $\delta_0 = 2^{-2n^2}$, and round each real and imaginary component after addition or multiplication to absolute error at most $R = 2^{-n^2}$. Then the total alias-output error is at most*

$$32\,2^{-n^2+3n} < 1/16 \qquad (n \geq 8).$$

These operations can be performed with polynomially many charged word operations per phase or arithmetic operation. This is not a proof of coverage or of the separate checksum estimator's precision.

Proof. There are at most 2^n summands in either half-map and at most 2^n dictionary keys: distinct dictionary classes are nonempty and disjoint subsets of the 2^n possible index sets. The Euclidean error of a complex rounding is at most $2R$. Thus either computed half-map sum differs from its

exact value by at most $2^n\delta_0 + 2^{n+1}R \le 3\,2^nR =: e$. The exact sum has modulus at most 2^n, and its approximation has modulus at most 2^{n+1} for $n \ge 2$.

In one class product $\Phi_L\Phi_R\chi(Z)$, replacing the factors by their approximations changes the value by at most

$$e2^n + 2^{n+1}e + 2^{2n+2}\delta_0 \le 13\,2^{2n}R.$$

Two rounded multiplications contribute at most $6R$, because the approximate unit phase has modulus at most two. Hence a class term has error at most $20\,2^{2n}R$. Summing at most 2^n terms and rounding the additions gives error at most $20\,2^{3n}R + 2^{n+1}R \le 32\,2^{3n}R$. The displayed strict $1/16$ bound holds already at $n = 8$ and improves thereafter. All estimates are deterministic and uniform over the retained phases.

For an explicit effective phase evaluation, first sum the integer residues modulo P exactly, then approximate the bounded angle $2\pi k/P$. The identity $\pi/4 = \arctan(1/2) + \arctan(1/3)$ and the alternating arctangent series give a rational π enclosure to error 2^{-8n^2} using $16n^2$ terms. Rational division at the same precision gives an angle error $O(2^{-8n^2})$. Evaluate sine and cosine by their Taylor polynomials through degree $16n^2$, on $|x| \le 8$. The remainder is bounded by the exponential-series tail

$$\frac{8^{N+1}}{(N+1)!}\,\frac{1}{1-8/(N+2)}, \qquad N = 16n^2.$$

For $n \ge 8$, use $N! \ge (N/2)^{N/2}$ and $N \ge 1024$; this bound is at most $16\,2^{-24n^2}$. The angle error and Taylor errors together are strictly less than 2^{-2n^2} in complex norm, with ample slack.

All exact input phases have $O(n)$ bits because the published prime interval has $\log_2 P = O(n)$. Real and imaginary intermediate magnitudes are at most $2^{O(n)}$, even when summing all dictionary terms. Keeping n^2 fractional bits therefore uses polynomially many bits, as do the rational Taylor calculations. Elementary multiword integer arithmetic implements each operation with polynomially many word operations, charging retained storage and rounding. No transcendental oracle or uncharged table is required. Finite smaller n can use the exact fallback already described. □

10 Fixed data, formalization and the allowable conclusion

The published tuple, all 43 filters and the exact expected 430 filter/region pairs are fixed throughout this phase. A kernel proof of rational comparisons must be distinguished from a proof that those rationals enclose actual real logarithms or entropy. Likewise a generic transfer theorem must be instantiated with the actual functions and coefficient records; naming an arbitrary quantity a moment is insufficient. The written identities in this supplement do not by themselves certify an implementation, a Lean theorem, or an integration output.

The internal work exponent is exactly $\beta_{\mathrm{work}} = .48767228002$ and the advertised exponent is $\alpha = .4876723$, with $\alpha - \beta_{\mathrm{work}} = 1.998 \cdot 10^{-8}$. If an algorithmic argument supplies the actual all-history bound $2^{\beta_{\mathrm{work}}n+O(n^{\cdot 7})}\,\mathrm{poly}(n+n^c)$ for each fixed c, then it is $O_c(2^{\alpha n})$: taking base-two logarithms of the extra factor gives $O(n^{\cdot 7}) + O_c(\log n) = o(n)$, eventually below $(\alpha - \beta_{\mathrm{work}})n$. This is an asymptotic implication, not a proof of the premise's resource accounting or a license to omit that premise.

Theorem 14 (The strengthened conditional implication)**.** *Retain the published fixed tuple and its exact scalar certificate. Assume the remaining source algorithmic claims identified in Section 9: the averaged-layer/alias coverage and cap analysis of Kapila Sections 3 and 5, the block-table/checksum probability analysis of OpenAI Section 4 as adapted in Kapila Section 6, and their finite-precision/full-program implementation claims with the published parameters, guard and slack.*

Under the explicit validated arithmetic trust of Section 5, the same single randomized word-RAM program halts on every input and is correct with probability at least 2/3 on every input. For every fixed positive integer c its time on every random execution with $b \le n^c$ is $O_c(2^{\cdot 4876723n})$.

The 430 original continuous moment inequalities are no longer additional unverified premises: they are supplied by the validated certificates. This remains a conditional algorithmic theorem, not a full Lean proof or an independently verified complete subset-sum algorithm.

Proof. Equation (5) gives exactly the old margins needed by the five released transfer families. The concrete transfer proofs here and the unchanged exact scalar certificate supply a finite margin 10^{-7}; the prescribed rounding loss is bounded by $2532 \cdot 10^{-12}$, leaving $9.7468 \cdot 10^{-8} > 0$. Section 4 supplies the needed effective finite-circle approximation. These statements do not invoke an estimator success conclusion as a scalar or analytic hypothesis.

For fixed tie weights, split according to LD, not according to an observed CONTINUE. In the complementary case the preliminary procedure answers exactly except with probability $O(1/n)$; under LD the remaining alias and checksum hypotheses give the required uniform accuracy with failure probability $o(1)$. Use the same retained phases in both quantities. Sound class summation and exact cancellation leave zero on a NO input. On a YES input the isolation proof gives a unique exact bin with probability at least 11/12, whose phase has modulus one. A combined approximation error less than 1/4 separates the two cases at modulus threshold 1/2. The total failure is at most $1/12 + o(1)$, hence below 1/3 above a fixed cutoff; finite exceptions use exact enumeration. The separate alias-arithmetic bound proves its local part of this accuracy claim, but the checksum precision and coverage parts are explicitly retained source hypotheses.

The stated implementation hypotheses concern the specified subroutines and their deterministic caps, not a renamed assertion of the final decision theorem. They give the all-history bound with exponent β_{work} above. The input guard $10^{12}\lceil \log_2(b+1) \rceil \le n$ holds eventually for every fixed $b \le n^c$; a guard failure invokes the exact fallback. For each fixed c those exceptions occur at only finitely many n, and their cost is absorbed into the constant C_c. The program and its cutoff/guard do not depend on c. Strict exponent absorption then gives the displayed ordinary-O bound. Every-history cost and eventual probability are used separately; an expected-time bound is insufficient. □

What has and has not been removed. Compared with the published assumptions, actual score regularity/bounds, effective finite-circle approximation, isolation and the preliminary LD split now have reconstructed reviewed written proofs. Sound alias insertion, class summation and its local arithmetic error have written proofs as well. The entire old-moment family is validated outside Lean. The unproved averaged-layer/alias coverage and cap chain, checksum estimator/precision chain and complete implementation/resource chain remain as precisely listed source hypotheses. The new fixed-certificate Lean development is assessed separately; this numerical result alone does not settle its formal obligations or imply any site status.

Only exact obligations supported by completed evidence may be removed from the proposed strengthened conditional statement. In particular, unresolved analytic hypotheses in a formal theorem remain hypotheses even when its axiom list is clean. The allowed ordinary Lean foundational axioms are reported separately from explicit mathematical premises. Unfinished or experimental modules cannot enter the accepted dependency chain by assertion, a custom axiom or a native-evaluation trust shortcut.

Provenance and assistance. The density and numerical-convolution framework is OpenAI's; the 43-filter and alias construction is Annanay Kapila's; the earlier transfer framework and finite-data formalization are Joshua Wharton's [2, 1, 3]. The present supplement reconstructs the specific

applicability and proofs stated here. The intended author is Yanni Bills, assisted by Researcher-Ed_Lorenz for source/proof analysis, Computational-Richardson for formal and validated numerical work, Reviewer-Procopius for adversarial internal review, and the Coordinator for integration. Internal agent review is not independent human peer review or official website verification. Nothing in this supplement authorizes publication, a release change or a verification-status request.

References

[1] Annanay Kapila. Subset sum in time $O(2^{0.4877n})$: a complete proof. `https://github.com/a-kapila/maths-proofs/blob/44c34961b56784c220a6e913cc068d57dc37c5a3/subset-sum/PROOF_0.4877.tex`, 2026. Pinned commit 44c34961b56784c220a6e913cc068d57dc37c5a3. Sections 2–8 and Appendices A–D inspected; source assertions are distinguished from reconstructed proofs and validated results.

[2] OpenAI. Subset sum in time $O(2^{0.49n})$. `https://github.com/openai/math/tree/fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb/preprints/Subset-Sum-in-Time-2-power-0-49n-October-4-2026`, 2026. Pinned commit fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb. Directly inspected build sections setup, filtering, numerical, checksum, aliases and implementation.

[3] Joshua Wharton. A conditional refinement of the subset sum exponent to 0.487676. `https://github.com/JoshuaQFEX/subset-sum-bound/blob/dcb002174cb1e06213e171c9f5ff6b063fdf2c40/paper.tex`, 2026. Paper pin dcb002174cb1e06213e171c9f5ff6b063fdf2c40. Separately inspected formalization at bed8e18f96db5955c1424abf3005b4fed4bd8d77; any reused formal theorem requires its own build and explicit-hypothesis audit.